

KADISON'S THEOREM ON LINEAR ORDER ISOMORPHISMS OF C^* -ALGEBRAS HOLDS WITHOUT THE ADDITIVITY ASSUMPTION

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ABSTRACT. In this short note, we prove that the additivity assumption in Kadison's celebrated result on the structure of linear order isomorphisms of C^* -algebras can be dropped.

In what follows, C^* -algebras are supposed to be unital. If $\mathcal{A}$ is a C^* -algebra, then $\mathcal{A}_s$ denotes the real-linear space of its selfadjoint elements, $\mathcal{A}^+$ stands for the positive semidefinite cone of $\mathcal{A}$ (i.e., the set of all elements of $\mathcal{A}_s$ with nonnegative spectrum) and $\mathcal{A}^{++}$ denotes the positive definite cone of $\mathcal{A}$ (i.e., the set of all elements of $\mathcal{A}_s$ with strictly positive spectrum). The usual (Löwner) order on $\mathcal{A}_s$ is defined as follows. For any $A, B \in \mathcal{A}_s$, we write $A \leq B$ if and only if $B - A \in \mathcal{A}^+$.

A celebrated result by Kadison [2] describes the structure of all linear order isomorphisms between the selfadjoint parts of C^* -algebras. It reads as follows.

Theorem 1. *Let $\mathcal{A}, \mathcal{B}$ be C^* -algebras, $\phi : \mathcal{A}_s \rightarrow \mathcal{B}_s$ be a linear order isomorphism, that is a bijective linear map with the property that for any $A, B \in \mathcal{A}_s$ we have*

$$A \leq B \iff \phi(A) \leq \phi(B).$$

Then there is an invertible element $T \in \mathcal{B}$, and a Jordan $$ -isomorphism $J : \mathcal{A} \rightarrow \mathcal{B}$ such that*

$$(1) \quad \phi(A) = TJ(A)T^*, \quad A \in \mathcal{A}_s.$$

Kadison obtained this result (in a slightly different but equivalent form) as a consequence of a so-called generalized Schwarz inequality for contractive positive linear maps, see Theorem 1 and Corollary 5 in [2].

A Jordan $*$ -isomorphism between the C^* -algebras $\mathcal{A}, \mathcal{B}$ is a bijective linear map $J : \mathcal{A} \rightarrow \mathcal{B}$ such that

$$J(AB + BA) = J(A)J(B) + J(B)J(A), \quad A, B \in \mathcal{A}$$

and

$$J(A^*) = J(A)^*, \quad A \in \mathcal{A}.$$

It is easy to see that any such map, when restricted to the space of selfadjoint elements, is a linear order isomorphism, hence the maps that appear in (1) above are all linear order isomorphisms between $\mathcal{A}_s$ and $\mathcal{B}_s$.

The aim of this note is to point out that the additivity assumption in Kadison's theorem is superfluous, we get it for free. As for the homogeneity, it is obvious that it cannot be dropped at the same time, just think of the case of the complex field $\mathbb{C}$ as a C^* -algebra. We have the following.

Theorem 2. *The additivity of the map ϕ in Kadison's theorem above can be dropped while still having the same structural description (1) of ϕ .*

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The proof has two ingredients. The first one comes from a former joint result by the first author, which describes the surjective Thompson isometries between the positive definite cones of C^* -algebras. The details are as follows. The Thompson metric d_T on $\mathcal{A}^{++}$ is given by

$$d_T(A, B) = \|\log(A^{1/2}B^{-1}A^{1/2})\|, \quad A, B \in \mathcal{A}^{++}.$$

The mentioned result, Theorem 9 in [1], says that if $\phi : \mathcal{A}^{++} \rightarrow \mathcal{B}^{++}$ is a surjective map which is a Thompson isometry, i.e., it satisfies

$$d_T(\phi(A), \phi(B)) = d_T(A, B), \quad A, B \in \mathcal{A}^{++},$$

then there is a Jordan $*$ -isomorphism $J : \mathcal{A} \rightarrow \mathcal{B}$, an invertible element $T \in \mathcal{B}$, and a central projection $P \in \mathcal{B}$ such that

$$\phi(A) = T(PJ(A) + (I - P)J(A^{-1}))T^*, \quad A \in \mathcal{A}^{++}.$$

This result has the following consequence, which has already been proved to be useful in the solutions of some other problems. We present it with proof for the sake of completeness.

Proposition 3. *Assume that $\phi : \mathcal{A}^{++} \rightarrow \mathcal{B}^{++}$ is a positive homogeneous order isomorphism, i.e., for any $A, B \in \mathcal{A}^{++}$, $t > 0$ we have*

$$\phi(tA) = t\phi(A), \quad \text{and} \quad A \leq B \iff \phi(A) \leq \phi(B).$$

Then we have a Jordan $$ -isomorphism $J : \mathcal{A} \rightarrow \mathcal{B}$ and an invertible element $T \in \mathcal{B}$ such that*

$$\phi(A) = TJ(A)T^*, \quad A \in \mathcal{A}^{++}.$$

Proof. Indeed, observe that $d_T(A, B)$ is the smallest nonnegative real number ϵ for which

$$-\epsilon I \leq \log(A^{1/2}B^{-1}A^{1/2}) \leq \epsilon I.$$

These inequalities are equivalent to

$$e^{-\epsilon}B \leq A \leq e^{\epsilon}B.$$

Since ϕ is a positive homogeneous order isomorphism, we have the equivalent inequalities

$$e^{-\epsilon}\phi(B) \leq \phi(A) \leq e^{\epsilon}\phi(B)$$

and then the ones

$$-\epsilon I \leq \log(\phi(A)^{1/2}\phi(B)^{-1}\phi(A)^{1/2}) \leq \epsilon I.$$

Therefore, we have that ϕ is a surjective Thompson isometry. By [1, Theorem 9], there exist a Jordan $*$ -isomorphism $J : \mathcal{A} \rightarrow \mathcal{B}$, an invertible element $T \in \mathcal{B}$, and a central projection $P \in \mathcal{B}$ such that

$$(2) \quad \phi(A) = T(PJ(A) + (I - P)J(A^{-1}))T^*, \quad A \in \mathcal{A}^{++}.$$

We show that $P = I$. For every $A \in \mathcal{A}_{++}$ and $t > 0$, positive homogeneity gives

$$\phi(tA) = t\phi(A).$$

Using (2), we obtain

$$\phi(tA) = T(tPJ(A) + t^{-1}(I - P)J(A^{-1}))T^*,$$

whereas

$$t\phi(A) = T(tPJ(A) + t(I - P)J(A^{-1}))T^*.$$

Subtracting yields

$$(t^{-1} - t) T(I - P)J(A^{-1})T^* = 0.$$

Choosing $t \neq 1$ and $A = I$, we get

$$T(I - P)T^* = 0,$$

hence $I - P = 0$, and (2) reduces to

$$\phi(A) = TJ(A)T^*, \quad A \in \mathcal{A}^{++}.$$

This proves the proposition. $\square$

The other ingredient of the proof of Theorem 2 is the following lemma in which we characterize the equality of selfadjoint elements with the help of the order and the positive semidefinite cone. For any $X \in \mathcal{A}_s$, denote $\mathcal{P}_X = \{A \in \mathcal{A}^+ : X \leq A\}$ and $\mathcal{N}_X = \{-A : A \in \mathcal{A}^+, -A \leq X\}$.

Lemma 4. *If $X, Y \in \mathcal{A}_s$ are such that $\mathcal{P}_X = \mathcal{P}_Y$ and $\mathcal{N}_X = \mathcal{N}_Y$, then it follows that $X = Y$.*

Proof. Let $X \in \mathcal{A}_s$ be given. We have the simple observation that the map $A \mapsto X - A$ maps $\mathcal{P}_X$ onto $\mathcal{N}_X$. After this, for any positive linear functional φ on $\mathcal{A}$, we have that

$$\sup_{N \in \mathcal{N}_X} \varphi(N) = \sup_{A \in \mathcal{P}_X} (\varphi(X) - \varphi(A)) = \varphi(X) - \inf_{A \in \mathcal{P}_X} \varphi(A)$$

which gives us

$$\varphi(X) = \inf_{A \in \mathcal{P}_X} \varphi(A) + \sup_{N \in \mathcal{N}_X} \varphi(N)$$

Therefore, we immediately obtain that $\varphi(X) = \varphi(Y)$ holds for any positive linear functional φ on $\mathcal{A}$. Since those functionals separate the points of $\mathcal{A}_s$, we conclude that $X = Y$. $\square$

Proof of Theorem 2. Let $\phi : \mathcal{A}_s \rightarrow \mathcal{B}_s$ be a homogeneous order isomorphism, i.e., assume that

$$\phi(tA) = t\phi(A)$$

and

$$A \leq B \iff \phi(A) \leq \phi(B)$$

hold for any $A, B \in \mathcal{A}_s$ and real number t .

Clearly, we have $\phi(0) = 0$. It follows that $\phi(\mathcal{A}^+) = \mathcal{B}^+$.

Apparently, a positive semidefinite element $A \in \mathcal{A}^+$ is positive definite if and only if for any $B \in \mathcal{A}^+$ there is a positive scalar t such that $B \leq tA$. Using this characterization, it follows that ϕ maps $\mathcal{A}^{++}$ onto $\mathcal{B}^{++}$. But it then follows from Proposition 3 that there is a Jordan *-isomorphism $J : \mathcal{A} \rightarrow \mathcal{B}$ and an invertible element $T \in \mathcal{B}$ such that $\phi(A) = TJ(A)T^*$, $A \in \mathcal{A}^{++}$.

Define $\psi(B) = J^{-1}(T^{-1}BT^{-1*})$, $B \in \mathcal{B}_s$, and consider the map $G = \psi \circ \phi$. This is a homogeneous order isomorphism on $\mathcal{A}_s$ which is the identity on $\mathcal{A}^{++}$ and we are done as soon as we prove that G is the identity everywhere.

First, G is the identity on $\mathcal{A}^+$. Indeed, if two positive semidefinite elements have the same positive definite majorants, then those two elements coincide. So, for any $A \in \mathcal{A}^+$ and $B \in \mathcal{A}^{++}$ we have

$$A \leq B \iff G(A) \leq G(B) = B$$

implying that $G(A) = A$. Therefore, G is the identity on $\mathcal{A}^+$ and the same is true on the set $-\mathcal{A}^+$.

Now, for an arbitrary $X \in \mathcal{A}_s$ we easily have that $\mathcal{P}_{G(X)} = \mathcal{P}_X$, $\mathcal{N}_{G(X)} = \mathcal{N}_X$. It follows from Lemma 4 that $G(X) = X$, $X \in \mathcal{A}_s$ and we are done. $\square$

The essence of the argument above can actually be used to verify the following general statement. For its formulation, we need the following definitions.

Definition 5. Let X be a real vector space and $C \subseteq X$ be a cone that is proper, meaning that $C \cap (-C) = \{0\}$, hence inducing a partial order $\leq$ via

$$x \leq y \iff y - x \in C.$$

An element $u \in C$ is called an order unit if for every $x \in X$ there exists $\lambda > 0$ such that

$$-\lambda u \leq x \leq \lambda u.$$

The pair (X, C) is called Archimedean if we further have that the implication

$$\forall \epsilon > 0 : x + \epsilon u \in C \implies x \in C$$

holds for any $x \in X$. Finally, the triple (X, C, u) is called an order unit space (see [3, §2]) if C is a proper cone in X , which is also generating in the sense that $X = C - C$, u is an order unit, and (X, C) is Archimedean.

Apparently, if $\mathcal{A}$ is a C^* -algebra, then $(\mathcal{A}_s, \mathcal{A}^+, 1)$ is an order unit space. Now, the general result reads as follows.

Theorem 6. Let (X, C, u) and (Y, K, v) be order unit spaces. Assume that the positive linear functionals on X (i.e., linear functionals that take nonnegative values on C) separate the points of X . If $\phi: X \rightarrow Y$ is a homogeneous order isomorphism, then ϕ is additive.

Obviously, this general result does not include any structural description of the maps in question (unlike Theorem 2), but points out their automatic additivity.

Proof. We present the sketch of the proof. Having the previous arguments in mind, one can easily make it complete.

First, since ϕ is homogeneous, we have $\phi(0) = 0$, hence ϕ maps C onto K .

Based on an old result by Schäffer [7], Lemmens, van Gaans, and van Imhoff pointed out in [3] (see the discussion after Theorem 5.3) that any nonnegative homogeneous order isomorphism between C and K is additive. It follows that $\phi|_C: C \rightarrow K$ is additive.

Now, define the map $F: X \rightarrow Y$ in the following way: for any $a, b \in C$, set

$$F(a - b) = \phi(a) - \phi(b).$$

By the homogeneity of ϕ and its additivity on C , one can easily check that F is a well-defined bijective linear order isomorphism from X onto Y which coincides with ϕ on C . Next, consider the map $F^{-1} \circ \phi$. It is a homogeneous order isomorphism on X which acts as the identity on C .

One can easily verify a straightforward variant of Lemma 4 for the setting of order unit spaces whose elements are separated by positive linear functionals and then use it to conclude that $F^{-1} \circ \phi$ is necessarily the identity on the whole space X . This implies that $\phi = F$, i.e., ϕ is additive. $\square$

Let us close with a few remarks. First, one can note that Schäffer's theorem, Theorem B in [7], which appears in [3] as Theorem 5.2, would give the additivity of positive homogeneous order isomorphisms between positive definite cones of C^* -algebras. That is true, but our Proposition 3, besides the additivity, also provides the precise structural description of such maps and then this extends further to all homogeneous order isomorphisms between the self-adjoint parts of C^* -algebras (Theorem 2). On the other hand, we believe that it is an interesting point to observe correspondences between positive homogeneous order isomorphisms of positive definite cones and Thompson isometries.

Second, we should point out that there have been serious investigations to obtain results that would guaranty the automatic linearity/affinity of order isomorphisms between positive cones or selfadjoint parts of operator algebras. In this respect, we refer to [4], [5], [6], and also to [3] in the context of general partially ordered vector spaces. Of course, as the simple example of the algebra $\mathbb{C}$ shows, to obtain such a result one needs to pose some extra conditions.

All in all, we believe that the general observation concerning Kadison's theorem in the setting of general C^* -algebras that we have made here deserves to be noted.

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